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# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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**SDC HACKATHON 2026**

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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | Chili Raccoon 7 |
| **Project / Solution Name** | FutureReady |
| **Name (Leader)** | Shiu Jun Ming |
| **Email & Phone (Leader)** | shiujm2007@gmail.com |
| **University Affiliation(s)** | Xiamen University Malaysia (XMUM) |
| **Live Prototype URL** | https://future-ready-workforce-shiujm2007.replit.app/ |
| **Code Repository URL (public / access granted to judges)** | https://github.com/JmShiu06/FutureReady-Workforce.git |

## 2. Architecture & Technology Stack

### 2.1 System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

FutureReady is a web app. React and TypeScript build the screens, and the Express API handles requests. Career matching compares selected skills with built-in career references; market pages use public Malaysian labour-market data. For AI explanations, the backend calls a Qwen3-4B model hosted on RunPod. CV files are read by the backend using PDF/DOCX text extraction and OCR. The raw CV file and full CV text are not sent to RunPod. The skill anlaysis feature sends selected skill names and calculated matches, gaps and learning-resource details after consent. Most profile and learning data is stored in browser localStorage. Job Search featureuses fictional sample jobs, not live vacancies.

**See appendix A for the system block diagram.**

### 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| TypeScript 5.9 | Main language used to write the frontend and backend app code. | Apache-2.0. |
| JavaScript | Used for web application code and JavaScript-based tooling. | ECMAScript standard. |
| Python utility scripts | Used for offline dataset checks; Python is not the app server runtime. | Python Software Foundation (PSF) License. |
| React 19 | Builds the interactive frontend screens. | MIT. |
| Express 5 | Handles backend routes and API requests. | MIT. |
| Node.js 20 | Runs the backend JavaScript code. | MIT. |
| Vite 7 | Builds the frontend and supports the development/build process. | MIT. |

| Name & Version | Purpose in Project | License |
|---|---|---|
| Tailwind CSS 4 | Provides utility classes used to style the pages. | MIT. |
| React DOM 19 | Connects the React interface to the browser page. | MIT. |
| Wouter 3 | Handles navigation between pages. | MIT. |
| Radix UI | Provides reusable interface components. | MIT; individual package licences apply. |
| Zod | Checks the shape of data used in API requests and responses. | MIT. |
| TanStack Query | Fetches and manages server data in the frontend. | MIT. |
| Drizzle ORM | Database/schema code is present, but active database use was not found in the current API routes. | Apache-2.0. |
| Multer 2 | Receives CV uploads in backend memory for processing. | MIT. |
| pdf-parse 2.4.5 | Extracts text from PDF CVs. | Apache-2.0. |
| pdfjs-dist 5.4.296 | Supports PDF processing and text extraction. | Apache-2.0. |
| Mammoth 1.13.0 | Extracts text from DOCX CVs. | BSD-2-Clause. |
| Tesseract.js 7.0.0 | Uses OCR to read text from CV images. | Apache-2.0. |
| @tesseract.js-data/eng 1.0.0 | English-language data used by Tesseract OCR. | MIT. |

## 3. APIs & Cloud Services

### 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| RunPod HTTPS API (vLLM / chat completions) | Runs Qwen 3-4B to explain career matches and skill gaps. | Sponsor-provided |
| OpenDOSM API | Gets the monthly labour force survey data for market pages | Free |
| Google Fonts API | Loads the DM Sans, Manrope and IBM Plex Mono fonts | Free |

### 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit | Hosts the web app and backend when deployed. | Core Plan / Asia |
| RunPod GPU Pod Service | Runs the Qwen model for AI explanation and analysis. | Basic Plan (25 USD) / Canada - Montreal (Data Center 1) |
| GitHub | Stores the project source code for the judging section | Basic Plan / None |

## 4. AI Tools, Models & Datasets

### 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| Qwen 3-4B (Qwen/Qwen3-4B) | Qwen / Hugging Face | Pre-trained | Apache-2.0 |

### 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| AI_Job_Market_Dataset.csv (job-title list) | https://www.kaggle.com/code/nuhmanpk/ai-job-market-analysis/input | Public | CC BY 4.0 | N |
| Most In-Demand Job Skills of 2026 (Qarera; career_roles.json) | https://doi.org/10.5281/zenodo.21204423 | Public | CC BY 4.0 | N |
| DOSM Job Market Insights Q1/Q2 2026 and Dec 2025 Labour Force Survey | DOSM release pages/workbooks; https://storage.dosm.gov.my/labour/lfs_month_2025-12_en.pdf | Public | Malaysia Government Open Data Terms of Use 1.0; attribution included | N |
| OpenDOSM Monthly Labour Force Survey (lfs_month) | https://open.dosm.gov.my/data-catalogue/lfs_month | Public | CC BY 4.0 | N |
| O*NET 31.0 skill ratings | https://www.onetcenter.org/db_releases.html | Public | CC BY 4.0 | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| MyMahir Malaysia Critical Occupations List (MyCOL) 2024/2025 | https://www.talentcorp.com.my/images/uploads/publication/204/MyMahir-Malaysia-Critical-Occupations-List-MyCOL20242025-1773385490.pdf | Public | Malaysia Government Open Data Terms of Use 1.0 | N |
| Self-made profiles, skill examples | Created by the team for prototype | Synthetic | None | N |
| Learning Resource and Certification Catalogue | Team-curated catalogue using official provider websites, including Cisco Networking Academy, Microsoft Learn, AWS Skill Builder and OpenEDG Python Institute | Self-collected | Individual provider terms apply | N |

### 4.3 AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✓ Yes ☐ No

If yes, list them below:

| **Tool & Version** | **Used For** | **Extent of Use (minor / moderate / major)** |
|---|---|---|
| ChatGPT (OpenAI) | Provide guidance related to features. | Minor |
| Replit Agent | Help write and improve code, tests and interface changes. | Major |
| Claude (Anthropic) | Helped with planning, debugging, prompts, source checks, documentation and design ideas. | Major |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

We reviewed the code and design changes suggested by ChatGPT and Replit Agent to make sure they suited our project. We tested the website's main features and ran the available tests and build checks. We also checked the results and made adjustments when needed. Our team is responsible for the final code, project functionality, and the accuracy and proper use of the information in our application.

## 5. Sponsor-Provided Resources

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

| **Sponsor** | **Resource (API / credits / platform / data)** | **How It Was Used in the Project** |
|---|---|---|
| Replit | 20 USD Credits | Develop and host our website and backend. |
| RunPod | 25 USD Credits | Used to run the Qwen3-4B AI model for career and skill explanations. |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

Replit provided a convenient platform for our team to develop and host our project, while RunPod provided the computing resources needed to run our Qwen3-4B AI model. Their support played an important role in helping us build and demonstrate our AI-powered career guidance prototype. We are grateful for these resources, which helped make our project possible within the hackathon.

## Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

When a user uploads a CV, the file is sent to our Replit-hosted backend for temporary text extraction, OCR and skill detection. The application processes it in memory and does not save the uploaded file or send the file or full extracted text to RunPod. The detected name and CV review remain in a temporary session; reviewed skills are saved to the browser profile only after the user confirms them. Profile skills, proficiency levels, preferences and learning progress may be stored in browser localStorage until cleared. With explicit consent, RunPod receives selected skill names, related career matches and skill gaps, and relevant catalogue/reference data for AI explanations. Other AI explanations use fictional examples and reviewed public information. The application backend does not cache or log CV contents or AI prompts and responses.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

Qwen3-4B may generate inaccurate or overly general explanations. To reduce this risk, users must give consent before AI analysis, and only selected skill and language names are sent to the AI. Career matches, scores and learning-resource suggestions are based on existing reference data rather than being freely generated by the model. CV text extraction may also miss information. Career scores show skill overlap, not hiring probability or proof of ability. Labour-market data covers broad occupation groups and may change between reporting periods; it does not represent all job vacancies. Synthetic examples are clearly labelled. Current tests use sample data and a mock AI service, so live AI responses and all CV extraction cases have not been fully verified.

## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

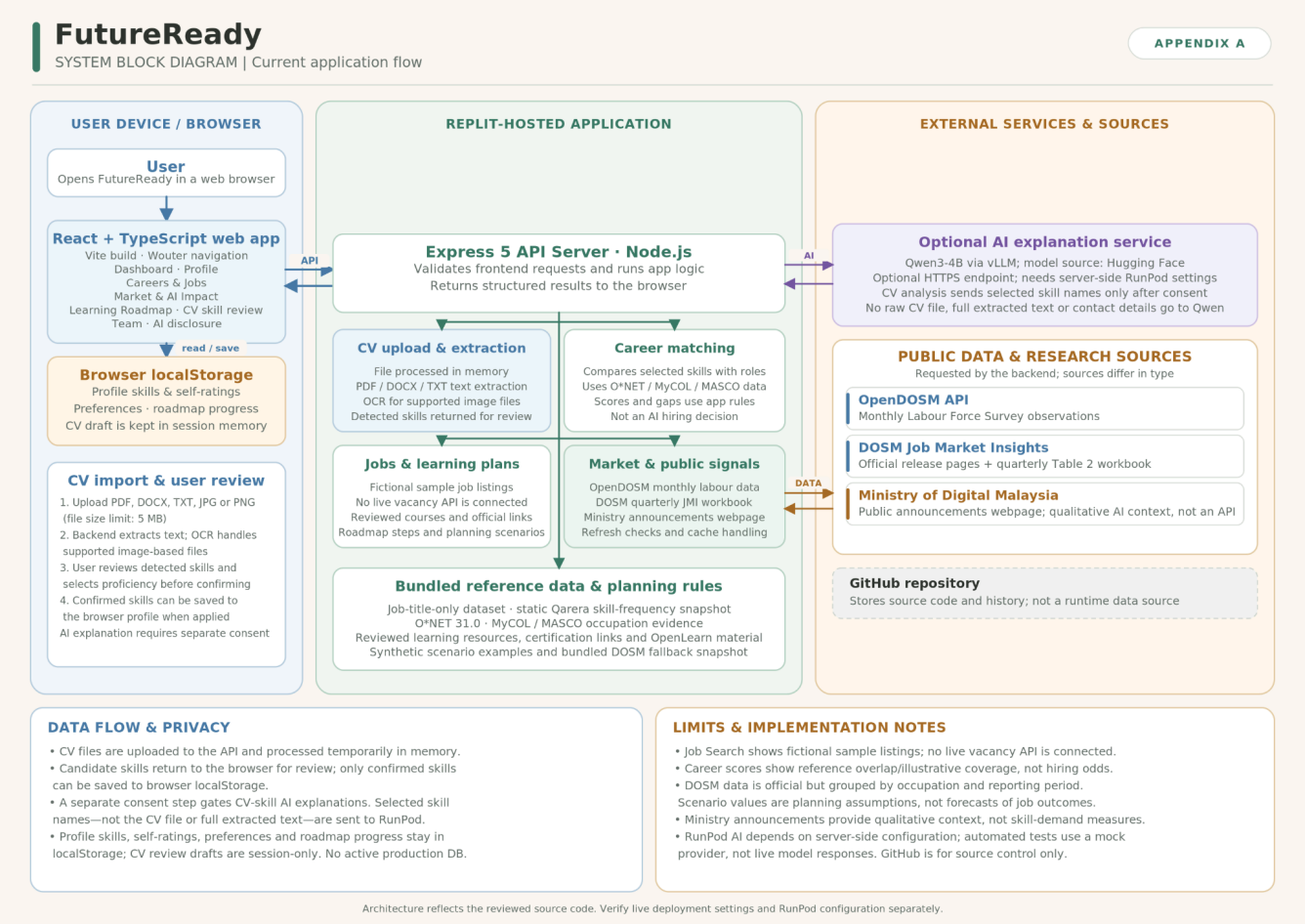
## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



24th October 2026

# Appendix


FutureReady
SYSTEM BLOCK DIAGRAM | Current application flow
APPENDIX A
USER DEVICE / BROWSER
User
Opens FutureReady in a web browser
React + TypeScript web app
Vite build · Wouter navigation
Dashboard · Profile
Careers & Jobs
Market & AI Impact
Learning Roadmap · CV skill review
Team · AI disclosure
read / save
Browser localStorage
Profile skills & self-ratings
Preferences · roadmap progress
CV draft is kept in session memory
CV import & user review
1. Upload PDF, DOCX, TXT, JPG or PNG (file size limit: 5 MB)
2. Backend extracts text; OCR handles supported image-based files
3. User reviews detected skills and selects proficiency before confirming
4. Confirmed skills can be saved to the browser profile when applied
AI explanation requires separate consent
API
REPLIT-HOSTED APPLICATION
Express 5 API Server · Node.js
Validates frontend requests and runs app logic
Returns structured results to the browser
CV upload & extraction
File processed in memory
PDF / DOCX / TXT text extraction
OCR for supported image files
Detected skills returned for review
Career matching
Compares selected skills with roles
Uses O*NET / MyCOL / MASCO data
Scores and gaps use app rules
Not an AI hiring decision
Jobs & learning plans
Fictional sample job listings
No live vacancy API is connected
Reviewed courses and official links
Roadmap steps and planning scenarios
Market & public signals
OpenDOSM monthly labour data
DOSM quarterly JMI workbook
Ministry announcements webpage
Refresh checks and cache handling
Bundled reference data & planning rules
Job-title-only dataset · static Qarera skill-frequency snapshot
O*NET 31.0 · MyCOL / MASCO occupation evidence
Reviewed learning resources, certification links and OpenLearn material
Synthetic scenario examples and bundled DOSM fallback snapshot
AI
DATA
EXTERNAL SERVICES & SOURCES
Optional AI explanation service
Qwen3-4B via vLLM; model source: Hugging Face
Optional HTTPS endpoint; needs server-side RunPod settings
CV analysis sends selected skill names only after consent
No raw CV file, full extracted text or contact details go to Qwen
PUBLIC DATA & RESEARCH SOURCES
Requested by the backend; sources differ in type
OpenDOSM API
Monthly Labour Force Survey observations
DOSM Job Market Insights
Official release pages + quarterly Table 2 workbook
Ministry of Digital Malaysia
Public announcements webpage; qualitative AI context, not an API
GitHub repository
Stores source code and history; not a runtime data source
DATA FLOW & PRIVACY
• CV files are uploaded to the API and processed temporarily in memory.
• Candidate skills return to the browser for review; only confirmed skills can be saved to browser localStorage.
• A separate consent step gates CV-skill AI explanations. Selected skill names—not the CV file or full extracted text—are sent to RunPod.
• Profile skills, self-ratings, preferences and roadmap progress stay in localStorage; CV review drafts are session-only. No active production DB.
LIMITS & IMPLEMENTATION NOTES
• Job Search shows fictional sample listings; no live vacancy API is connected.
• Career scores show reference overlap/illustrative coverage, not hiring odds.
• DOSM data is official but grouped by occupation and reporting period. Scenario values are planning assumptions, not forecasts of job outcomes.
• Ministry announcements provide qualitative context, not skill-demand measures.
• RunPod AI depends on server-side configuration; automated tests use a mock provider, not live model responses. GitHub is for source control only.
Architecture reflects the reviewed source code. Verify live deployment settings and RunPod configuration separately.